# MSWAGAN: Multispectral Remote Sensing Image Super-Resolution Based on Multiscale Window Attention Transformer

ChunYang Wang, Xian Zhang, Wei Yang, GaiGe Wang, *Member, IEEE*, XingWang Li, *Senior Member, IEEE*, JianLong Wang, and BiBo Lu

***Abstract*— Remote sensing image super-resolution (RSISR) techniques play a crucial role in various remote sensing applications. However, deep learning-based methods applied to RSISR encounter difficulties in learning complex features of remote sensing images and modeling long-term correlations between pixels. This study proposes a multiscale sliding window attention generation adversarial network (MSWAGAN), which combines the advantages of convolutional neural networks (CNNs) and transformers to overcome these limitations. The MSWAGAN consists of three main parts. In the shallow feature extraction part, CNN is used to extract shallow features from remote sensing images. The deep feature extraction part is divided into two stages. First, a multiscale sliding window attention (MSWA) is designed to replace the multihead attention (MHA) in the transformer. MSWA can learn local multiscale complex features of remote sensing images without increasing the number of parameters in MHA. Then, the transformer is utilized to learn global image features and model the long-range correlations between pixels. The image reconstruction part utilizes subpixel convolution for feature upsampling. Furthermore, in order to extend the application of super-resolution (SR) remote sensing images, a cross-sensor real multispectral RSISR dataset consisting of Landsat-8 (L8) and Sentinel-2 (S2) images was constructed, and a series of experiments to improve the spatial resolution of L8 images from 30 to 10 m in B, G, R, and near-infrared (NIR) bands were conducted. Experimental results demonstrate that our method outperforms some of the latest SR methods.**

Manuscript received 23 November 2023; revised 9 February 2024; accepted 13 March 2024. Date of publication 8 April 2024; date of current version 17 April 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFC3010701, in part by the Chunhui Program Cooperative Research Project of the Chinese Ministry of Education under Grant HZKY20220279, in part by the Henan Provincial Science and Technology Research Project under Grant 232102211019 and Grant 222102210131, and in part by the Key Research Project Fund of the Institution of Higher Education in Henan Province under Grant 23A520029. *(Corresponding author: Wei Yang.)*

ChunYang Wang is with the College of Computer Science and Technology and the School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo 454000, China (e-mail: wcy@hpu.edu.cn).

Xian Zhang is with the School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo 454000, China (e-mail: wjyts@home.hpu.edu.cn).

Wei Yang is with the Center for Environmental Remote Sensing, Chiba University, Chiba 263-8522, Japan (e-mail: yangwei@chiba-u.jp).

GaiGe Wang is with the School of Computer Science and Technology, Ocean University of China, Qingdao 266100, China (e-mail: wgg@ouc.edu.cn).

XingWang Li is with the School of Physics and Electronic Information Engineering, Henan Polytechnic University, Jiaozuo 454000, China (e-mail: lixingwangbupt@gmail.com).

JianLong Wang and BiBo Lu are with the College of Computer Science and Technology, Henan Polytechnic University, Jiaozuo 454000, China (e-mail: wangjianlong24@hpu.edu.cn; lubibo@hpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3385752

***Index Terms*— Deep learning, generating adversarial networks, multispectral remote sensing images, super-resolution (SR), transformer.**

## I. Introduction

SATELLITE technology advances have made remote sensing image data increasingly available and useful across various fields, such as disaster monitoring, agriculture, weather forecasting, and target detection [1], [2], [3], [4]. Remote sensing applications that rely on spatial information can benefit from high-resolution (HR) images, which provide clearer and richer texture information, leading to better results. However, obtaining high spatial resolution RS images remains challenging and expensive, given the limitations of optical sensors. Hence, modeling techniques have been employed by researchers to enhance the spatial resolution of RS images. Early methods such as Pansharp [5] and interpolation [6] were often used. However, Pansharp requires an HR panchromatic band sensor, while the interpolation method yields more ambiguous results. Other methods based on sparse representation [7] and convex projection [8] exist. However, these methods are too complicated and require a vast amount of a priori knowledge for implementation in remote sensing images with different sensors.

With advancements in computer technology, deep learning-based methods, particularly convolutional neural networks (CNNs), have been widely adopted in various research fields, including image super-resolution (SR). SRCNN [9] was the first to apply CNN to image SR tasks, achieving remarkable performance beyond traditional methods, and confirming the feasibility of this approach in image SR. Meanwhile, SRCNN has also been applied to RSISR [10]. However, remote sensing images differ from ordinary optical images, with complex features and scale differences between features. To address these challenges, researchers have designed new structures such as residual [11], [12], [13], multiscale [14], [15], [16], and attention-based [17], [18], [19] models to enhance CNN performance by building deeper networks and combining global–local contextual information of images. Despite the significant progress achieved using CNN, it is limited by its inability to handle the long-term dependencies of pixels, which hinders further improvement of CNN performance. To address this shortcoming, this study proposes an approach that integrates CNN and transformers, where the CNN is

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

employed to extract low-level features, and the transformer is utilized to capture high-level features as well as model the long-term dependencies among pixels.

The transformer, originally proposed in natural language processing (NLP), has become a backbone of the field due to its ability to operate on input data in parallel and its powerful long-term relational learning capability [20]. Recently, transformer-based methods [21], [22], [23], [24], [25] have been used in image SR tasks, with CNNs used for shallow feature extraction, while transformer blocks are connected with residuals or dense residuals to increase network depth and extract deeper features. This approach combines the advantages of CNN localization and transformer global learning, overcoming CNN's inability to model the long-term correlation between pixels. For remote sensing images, multiscale feature representation is essential [26], [27], [28] due to the complex types of features with varying scales. Several approaches have been proposed to introduce multiscale information into vision transformer (VIT) by changing the patch size [29], [30], improving the multihead attention (MHA) [31], [32], or adding additional branches [33], [34]. Nonetheless, such methods often result in an increased number of parameters, which consequently contributes to the slower training times of the already sizable VIT network. Inspired by [35], [36], [37], [38], and [39], this study suggests a multiscale sliding window attention (MSWA) mechanism, which introduces multiscale sliding windows into VIT standard self-attention. MSWA enables different heads to learn multiscale representations of different feature subspaces and computes sliding window-based multiscale attention without introducing additional parameters, thereby focusing on complex features of remote sensing images.

Generative adversarial networks (GANs) [40] have gained significant attention due to their ability to generate high-quality images in various fields [41], [42], [43]. SRGAN [43] was the first network to utilize GAN technology in image SR tasks. To generate images that align with human visual perception, SRGAN introduced a novel loss function that utilized a pretrained VGG network [44] to extract image features and improve the perceptual similarity between fake and real samples by optimizing the feature distances. However, the introduction of perceptual loss leads to artifacts in images [45], which can be exceptionally significant in the RSISR task. Many researchers have made substantial improvements in GAN networks for the RSISR task. These improvements include further edge sharpening of the generated SR images [46], attention mechanisms [47], gradient references [48], and utilization of remote sensing image characteristics [49]. Compared to traditional training methods, GANs can generate higher quality images due to progressive consistency. The gradient data from the GAN is not directly from real data but rather from the discriminator. In this study, we train the model using a GAN framework and a binary classification VIT network as the discriminator.

The objective of this article is to utilize deep learning techniques to enhance the spatial resolution of low-resolution (LR) RS images, thereby obtaining high-quality data for use in remote sensing applications. Specifically, we present a novel SR model for remote sensing images, MSWAGAN. To achieve optimal performance and accurate reconstruction from LR RS images, MSWAGAN's generator combines the powerful feature extraction capability of CNN and the long-term dependency modeling of transformers. Since remote sensing images feature complex feature types and different scales, we have designed a multiscale attention mechanism based on sliding windows to gain better insight into the multiscale image features. Most studies on RSISR methods employ the R, G, and B bands from natural image datasets. However, for some applications like land use classification, multispectral data yield better results. To extend the use of SR data, we have introduced a multiband remote sensing image SR dataset that includes publicly available L8 and S2 data collections. Our proposed method can SR four bands, including R, G, B, and near-infrared (NIR), and has broad applicability.

## II. Methods

### A. Overview of Proposed Method

Fig. 1 illustrates the structure of MSWAGAN proposed in this study. The generator is composed of three primary components: shallow feature extraction, deep feature extraction, and SR image reconstruction. The deep feature extraction stage consists of the MSWA stage and the VIT stage. Each stage contains multiple MSWA blocks or VIT blocks, a convolutional layer, and a residual connection.

Define the training dataset $\{I_i^{\mathrm{LR}} \in \mathbb{R}^{C_{\mathrm{in}}\times H\times W}, I_i^{\mathrm{HR}} \in \mathbb{R}^{C_{\mathrm{in}}\times H_r\times W_r}\}_{i=1}^{n}$ ($C_{\mathrm{in}}$ represents the number of bands, $H$ and $W$ represent the height and width of LR images, and $r$ represents the upsample factor). First, the shallow features $F_0 \in \mathbb{R}^{C\times H\times W}$ are extracted by a convolutional layer $H_{\mathrm{SF}}^{C\times K_s\times K_s}$ ($C$ represents the number of convolutional kernels and $K_s$ stands for the convolution kernel size) to map the input image onto a feature space with higher dimensions

$$F_0 = H_{\mathrm{SF}}\left(I^{\mathrm{LR}}\right). \tag{1}$$

The shallow features $F_0$ are partitioned into feature patches $F_{\mathrm{Patch}} \in \mathbb{R}^{C\times H_p\times W_p}(H_p = \frac{H}{P})$ of size $P \times P$. This partition is achieved using a convolutional layer with a convolutional kernel $H_{\mathrm{Patch}}^{C\times K_s\times K_s}$ with $K_s = P$

$$F_{\mathrm{Patch}} = H_{\mathrm{Patch}}(F_0). \tag{2}$$

Then, position encoding (PE) is performed for $F_{\mathrm{Patch}}$, because the VIT block calculates self-attention for all input feature patches in parallel, and the change of the order of feature patches will not affect the final attention calculation result. As shown in Fig. 2, this study using conditional positional encoding (CPE) [50], CPE uses a depth-wise convolutional [51] with kernel size $k$, ($k \geq 3$) and zero padding $\frac{k-1}{2}$ applied to 2-D image features $F_{\mathrm{Patch}}$ to generate the CPE, which can be represented as

$$\begin{aligned} \mathrm{PE} &= \mathrm{DWConv}(F_{\mathrm{Patch}}) \\ \hat{F}_{\mathrm{Patch}} &= \mathrm{PE} + F_{\mathrm{Patch}} \end{aligned} \tag{3}$$

where PE represents the location code generated by DWConv and $\hat{F}_{\mathrm{Patch}}$ represents the location information added to the $F_{\mathrm{Patch}}$. Since CPE is convolutionally generated, CPE is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

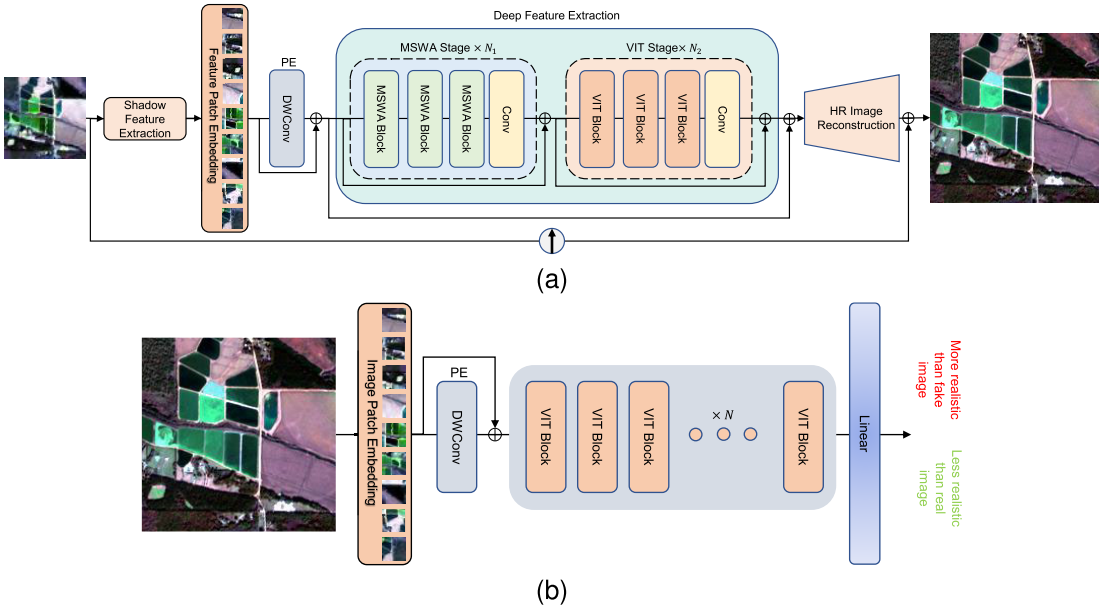



Fig. 1. General structure of the proposed method. (a) Generator. (b) Discriminator.

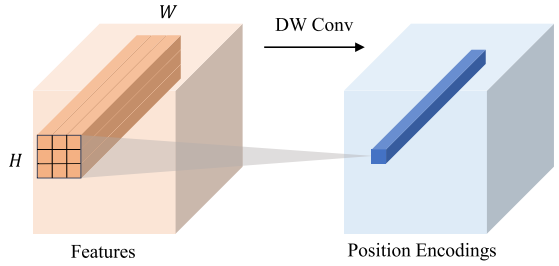



Fig. 2. CPE with DW Conv.

dynamically generated and conditioned based on the local neighborhood of the input features. This allows CPE to be applied to images of different resolutions without retraining. In addition, CPE can maintain the translation equivalence required in vision tasks, thus improving performance.

The depth features are extracted using $\hat{F}_{\text{Patch}}$ in the depth feature extraction module, and multiscale information is extracted using the MSWA block proposed in this study in the MSWA stage since the MSWA is localized based on sliding window calculation, the global depth information is extracted using the VIT stage after the MSWA stage to further model the long-term dependencies between global pixels. The depth feature can be expressed as

$$F_{\text{DF}} = \begin{cases} H_{\text{MSWAStage}}\left(\hat{F}_{\text{Patch}}\right) + \hat{F}_{\text{Patch}} \\ H_{\text{VITStage}}\left(\hat{F}_{\text{Patch}}\right) + \hat{F}_{\text{Patch}} \end{cases} \tag{4}$$

where $H_{\text{MSWAStage}}$ and $H_{\text{VITStage}}$ represent the feature extraction stages of the MSWA stage and VIT stage, respectively, and $F_{\text{DF}}$ represents the output features of the deep feature extraction module.

As shown in Fig. 3, the depth feature $F_{\text{DF}}$ undergoes two subpixel convolutions [52] to obtain the HR reconstructed features. The first subpixel convolution reconstructs the depth image patch feature as an LR feature, where $P$ denotes the

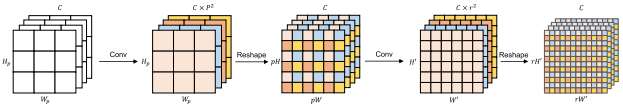



Fig. 3. Schematic of SR image reconstruction.

dimensions of the image patch. The second subpixel convolution achieves the SR feature by upsampling the image with a factor of $r$. Compared with interpolation, subpixel convolution can learn a more complex mapping of LR features to SR features, resulting in higher quality image reconstruction. Define the input feature is $F_{\text{LR}}$, the process of subpixel convolution can be expressed as

$$F_{\text{SR}} = \text{PS}(H_{\text{SP}}(F_{\text{LR}})) \tag{5}$$

where PS represents the periodic transformation operator, responsible for rearranging the feature tensor from $C \cdot r^2 \times H \times W$ to $C \times rH \times rW$. $H_{\text{SP}}$ refers to the convolution layer, which transforms the feature's channel number $C$ to $C \cdot r^2$. $H_{\text{SR}}$ denotes the SR image reconstruction module, representing the process of obtaining the final output SR image as follows:

$$I^{\text{SR}} = \text{UP}\left(I^{\text{LR}}\right) + H_{\text{SR}}\left(F_{\text{DF}} + \hat{F}_{\text{Patch}}\right) \tag{6}$$

where UP represents the upsampling operation, which upsamples the LR image to match the dimensions of the HR reconstructed feature map.

As shown in Fig. 1, this study utilizes a pure VIT classification model to undertake a binary classification task as a discriminator. The discriminator comprises $n$ VIT blocks , with the output being obtained through a linear classification head. In contrast to the traditional GAN approach of directly assessing the truthfulness of the input, this study employs a relative discriminator [53] to determine the probability that the input true image is more true than the false image and the input false image is more false than the true image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

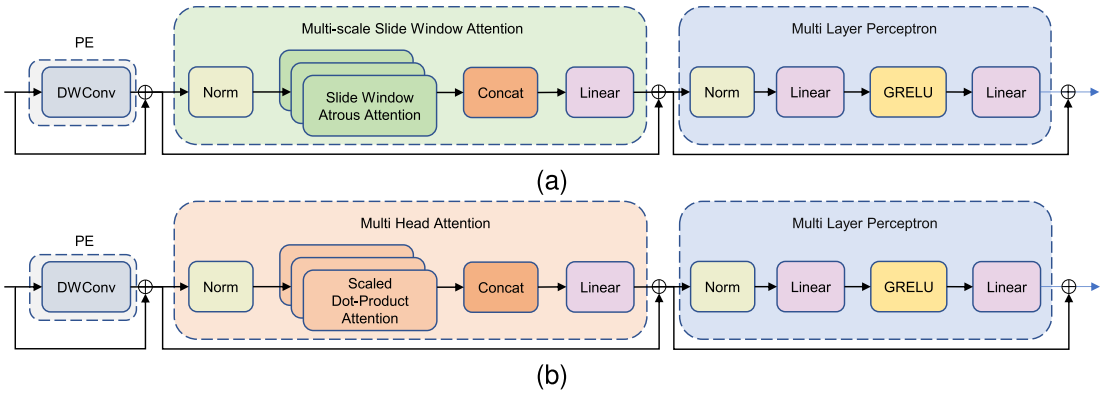



Fig. 4. (a) Multiscale slide window attention block. (b) VIT block.

### B. Multiscale Slide Window Attention Block

For RS images, the same region exhibits numerous multiscale features, and previous studies have shown the beneficial impact of multiscale structure on the image SR task [16], [27], [28]. However, in VIT, the dot product attention focuses solely on the feature information of RS images at a single scale, which fails to effectively capture the rich texture edge details present in remote sensing images. To leverage the multiscale information inherent in the images, this study enhances the global attention mechanism in VIT and proposes a new structure called MSWA block. Fig. 4 shows the detailed architecture of the MSWA block and compares it with the structure of the VIT block. The key distinction of the MSWA block from the VIT block is that instead of directly computing global attention directly on features, we compute self-attention within the window by employing sliding window slicing with different atrous rates for each head. With $F_{\text{input}}$ as the input, the MSWA block can be expressed as

$$\begin{aligned} Q &= \text{Linear}\big(\text{PE}\big(F_{\text{input}}\big)\big) \\ K &= \text{Linear}\big(\text{PE}\big(F_{\text{input}}\big)\big) \\ V &= \text{Linear}\big(\text{PE}\big(F_{\text{input}}\big)\big) \\ \text{head}_i &= \text{SWAA}((Q_i, K_i, V_i), r_i) \\ \text{MSWA} &= \text{Linear}(\text{Concat}(\text{head}_1, \text{head}_2, \cdots, \text{head}_i)) \\ \text{MSWA Block} &= \text{MLP}(\text{MSWA}). \end{aligned} \tag{7}$$

The $Q$, $K$, and $V$ matrices are obtained by applying three different linear transformations to the $F_{\text{input}}$. Subsequently, these matrices are divided equally into $i$ heads based on the channel dimension, with the $i$th head corresponding to $Q_i$, $K_i$, and $V_i$. The sliding window attention is then calculated for each head. Here, $r_i$ denotes the atrous rate of $\text{head}_i$. The different heads are combined together to yield the multiscale window attention. SWAA represents the abbreviation for sliding window atrous attention, which is the attention mechanism proposed in our work. Consequently, the MSWA block is constructed by utilizing a multilayer perceptron. Distinct from the transformer, the MSWA block positions the LayerNorm [54] at the start of the sublayer, rather than the end. This helps in facilitating the training process of the network. The PE follows the same approach as depicted in Fig. 2.

### C. Slide Window Atrous Attention

As shown in Fig. 5, for the query vector $q_{i,j}$ at the position of the Query matrix $(i, j)$, scale dot product attention computes the self-attention at the global scale. In contrast, the sliding window atrous attention (SWAA) proposed in this study computes the self-attention in a sliding window centered on $(i, j)$ and controls the size of the window as $w_r \times w_r$ by setting different void rates $r$. In this study, the value of $r$ was set to 1, 2, and 3 in the deep feature extraction module, and the convolutional kernel size of $3 \times 3$ for dilated convolutions. so that $q_{i,j}$ can perform multiscale queries and obtain multiscale representations of the corresponding position vectors of the output feature maps. Assuming that given an input feature map $F_{\text{in}}$, its $Q$, $K$, and $V$ matrices are obtained after linear transformation, the output feature vector $o_{i,j}$ at position $(i, j)$ of the input feature $F_{\text{in}}$ can be expressed as

$$o_{i,j} = \text{SWAA}\big(q_{i,j}, K_r, V_r\big) = \text{Softmax}\left(\frac{q_{i,j} K_r^T}{\sqrt{d_k}}\right) V_r \tag{8}$$

where $\sqrt{d_k}$ represents the scaling factor and $d_k$ denotes the default feature length of $K_r$. Softmax normalizes the matrix dot product to $[0, 1]$ in order to obtain the attention score. SWAA performs a self-attention operation on the query vector at all positions by using sliding windows. $K_r$ and $V_r$ denote sliding windows of different sizes on the $K$ and $V$ matrices. The proposed SWAA in this study incorporates both multiscale characteristics and localization, thereby addressing the limitations of VIT, which is limited to single-scale feature learning. Moreover, it effectively captures complex multiscale features of RS images and models long-term dependencies.

### D. Loss Function and Metric

This study combines pixel loss, perceptual loss, and relative adversarial loss to optimize the generator and improve the quality of SR images. The pixel loss is the $L1$ loss, which optimizes the SR image by minimizing the absolute difference between pixels. $L1$ pixel loss is known for its superior robustness compared to $L2$ loss. Mathematically, the pixel loss can be represented as follows:

$$\mathcal{L}_{\text{pixel}} = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} \left| G\big(I^{\text{IR}}\big)_{x,y} - I^{\text{HR}}_{x,y} \right| \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

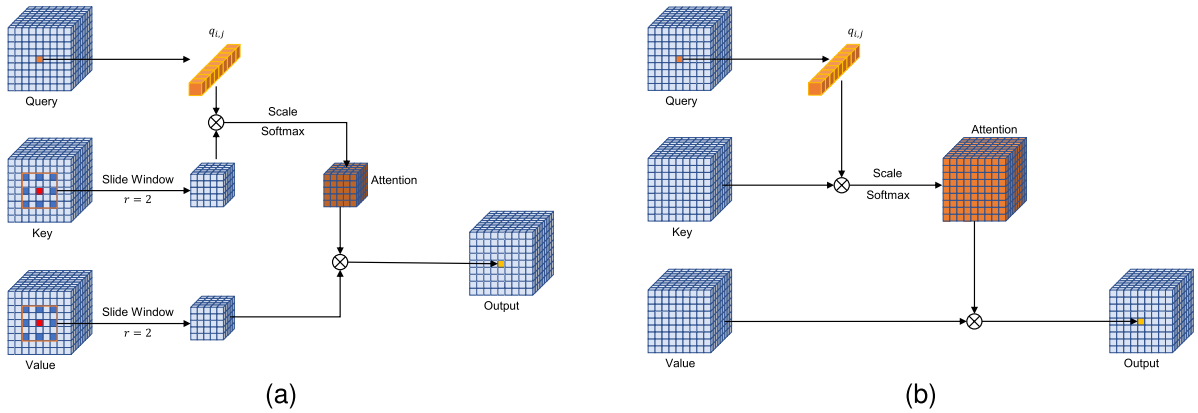



Fig. 5. Comparison of sliding window atrous attention and scale dot product attention. (a) Sliding window atrous attention. (b) Scale dot product attention.

where $H$ and $W$ denote the height and width of the image, respectively. $x$ and $y$ represent the number of rows in which the pixels are located and $G$ represents the generator. Although the SR image based on pixel loss optimization has a high peak PSNR value, the visual quality is not high due to the absence of high-frequency content.

SRGAN [43] proposed perceptual loss. The perceptual loss extracts the feature maps of SR and HR images, which contain rich spatial and semantic information, by a pretrained VGG [44] network, and then calculates the pixel distance between SR-HR feature maps as the perceptual loss. The perceptual loss optimizes the network in the high-dimensional perceptual space, which can weaken the effects caused by different angles of the ground objects and different lighting conditions. The perceptual loss can be expressed as

$$\mathcal{L}_{\text{vgg}} = \frac{1}{H_{i,j} \times W_{i,j}} \sum_{x=1}^{H_{i,j}} \sum_{y=1}^{W_{i,j}} \left| \Phi_{i,j}\left(G\left(I^{\text{IR}}\right)_{x,y}\right) - \Phi_{i,j}\left(I^{\text{HR}}_{x,y}\right) \right| \tag{10}$$

where $\phi_{i,j}$ denotes the feature map obtained from the $i$th convolution preceding the $j$th max-pooling layer within the VGG19 network after activation. $H_{i,j}$ and $W_{i,j}$ correspond to the height and width of the respective feature maps. Fig. 6 shows the visualization of the feature maps obtained from the initial convolution layer preceding the first five max-pooling layers.

When $j = 5$, the feature maps extracted by VGG19 may have negative values and the features are already highly sparse, rendering the 36th layer feature maps unnecessary. In this study, the VGG19 network's multilayer features are utilized to calculate the perceptual loss, enabling the comprehensive utilization of spatial information from low-level feature maps and ssemantic information from high-level features. The final perceptual loss is obtained by taking the weighted sum of perceptual losses from different layers. However, it is worth noting that utilizing multilayer features for calculating the perceptual loss increases memory usage and slows down training, although it does not affect the network during inference.

Regarding adversarial training, this study employs a relative average discriminator (RaGAN) [53]. Unlike the standard GAN, which aims to increase the probability of fake data being perceived as real while decreasing the probability of true data being recognized, RaGAN focuses on comparing the relative differences between real and generated data. This distinction enhances the discriminator's performance, reduces training instability commonly observed in standard GANs, and guides the generative network to generate more realistic images. The relative GAN loss can be expressed as

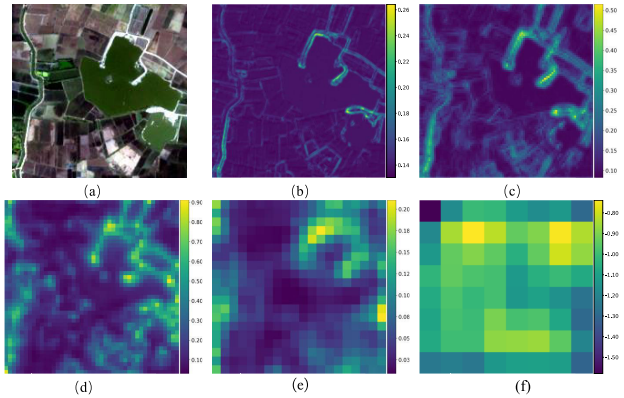



Fig. 6. VGG19 feature map visualization. (a) RS Image. (b) $j = 1$, $i = 1$. (c) $j = 2$, $i = 1$. (d) $j = 3$, $i = 1$. (e) $j = 4$, $i = 1$. (f) $j = 5$, $i = 1$.

$$\begin{aligned}\mathcal{L}_{\text{RaGAN}} = &-\mathbb{E}_{I^{\text{HR}} \sim \mathbb{P}\left(I^{\text{HR}}\right)}\left[\log\left(1 - D_{\text{RaGAN}}\left(I^{\text{HR}}, G\left(I^{\text{LR}}\right)\right)\right)\right] \\ &- \mathbb{E}_{G\left(I^{\text{IR}}\right) \sim \mathbb{P}\left(G\left(I^{\text{IR}}\right)\right)}\left[\log\left(D_{\text{RaGAN}}\left(G\left(I^{\text{LR}}\right), I^{\text{HR}}\right)\right)\right]\end{aligned} \tag{11}$$

where $G$ represents the generator, $G(I^{\text{LR}})$ represents the SR image generated by the generator, and $I^{\text{HR}}$ is the corresponding true HR image. $I^{\text{HR}} \sim \mathbb{P}(I^{\text{HR}})$ represents the distribution pattern of the real data and $G(I^{\text{IR}}) \sim \mathbb{P}(G(I^{\text{IR}}))$ represents the data distribution of the SR image. The complete loss is the weighted accumulation of the three losses, which can be expressed as

$$\mathcal{L} = \lambda \mathcal{L}_{\text{pixel}} + \beta \mathcal{L}_{\text{vgg}} + \gamma \mathcal{L}_{\text{RaGAN}} \tag{12}$$

where $\lambda$, $\beta$, and $\gamma$ represent the weight factors of loss, respectively, and are set to 1.0, 0.001, and 0.05.

In this study, peak signal-to-noise ratio (PSNR), structure similarity index measure (SSIM) [55], learned perceptual image patch similarity (LPIPS) [56], spectral angle mapper

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

(SAM) [57], and local standard deviation (LSD) are used to measure the quality of SR images. PSNR is a widely used image quality evaluation metric based on the difference between corresponding pixels. Minimizing pixel loss is equivalent to maximizing PSNR. Higher PNSR indicates higher quality of SR images, and it can be described as

$$\text{mse} = \frac{\sum |G(I^{\text{LR}}) - I^{\text{HR}}|^2}{H \times W}$$
$$\text{PSNR} = 20 \times \log_{10} \frac{\max(G(I^{\text{LR}}))}{\text{mse}(G(I^{\text{LR}}), I^{\text{HR}})} \tag{13}$$

where $H$ and $W$ represent the height and width of the SR image, mse denotes the mean square error between the SR image and the HR image, and max indicates the maximum possible pixel value of the image.

However, the PSNR metric does not consider the perceptual similarity as perceived by the human eye. Consequently, models that optimize pixel loss may generate visually unappealing images, despite achieving a high PSNR. On the other hand, SSIM is a perceptual evaluation measure that evaluates image quality in terms of brightness, contrast, and structure. This evaluation method closely aligns with the human eye's intuitive perception of images. The SSIM can be expressed as

$$\text{SSIM} = \frac{\left(2\mu_{G(I^{\text{LR}})}\mu_{I^{\text{HR}}} + C_1\right)\left(2\sigma_{G(I^{\text{LR}})I^{\text{HR}}} + C_2\right)}{\left(\mu^2_{G(I^{\text{LR}})} + \mu^2_{I^{\text{HR}}} + C_1\right)\left(\sigma^2_{G(I^{\text{LR}})} + \sigma^2_{I^{\text{HR}}} + C_2\right)} \tag{14}$$

where $\mu$ and $\sigma$ represent the mean and variance of the image, respectively. $C_1$ and $C_2$ are constants, defined as $k_1 L$ and $k_2 L$, respectively. $L$ represents the dynamic range of the image pixel values. $k_1$ and $k_2$ are usually set to 0.01 and 0.03, respectively.

LPIPS employs a pretrained neural network to extract deep-level feature maps from both SR and HR images, measuring the perceptual similarity between images by computing the $L_2$ distance between these feature maps. The LPIPS values range from [0, 1], with a lower value indicating a closer perceptual distance between the images. LPIPS can be expressed as

$$\text{LPIPS} = \frac{1}{H_l \times W_l} \| w_l \odot \left(\psi_l\left(G\left(I^{\text{LR}}\right)\right), \psi_l\left(I^{\text{HR}}\right)\right) \|_2^2 \tag{15}$$

where $\psi_l$ represents the $l$th layer of the pretrained network, $w_l$ denotes a vector used for channel-wise scaling of the feature map, and $H_l$ and $W_l$ signify the height and width of the output feature map at the $l$th layer of the pretrained network.

SAM is utilized to quantify the spectral similarity between multispectral remote sensing images. The lower SAM value indicates a higher spectral similarity. SAM can be expressed as

$$\text{SAM} = \frac{1}{H \times W} \arccos\left(\frac{G\left(I^{\text{LR}}\right)\left(I^{\text{HR}}\right)^{\text{T}}}{\|I^{\text{HR}}\|_2 \|G\left(I^{\text{LR}}\right)\|_2}\right) \tag{16}$$

where $(I^{\text{HR}})^{\text{T}}$ represents the transpose of $I^{\text{HR}}$.

The gray value of the edge region of features in remote sensing images typically undergoes drastic changes. LSD is capable of reflecting the extent of these changes. Therefore, LSD can detect the edge details of the image and measure the impact of the SR image reconstruction on the original image's details. LSD can be expressed as

$$\text{LSD} = \sqrt{\frac{1}{K^2} \sum_{i=1}^{k} \sum_{j=1}^{k} \left(I_{i,j} - M_k\right)^2} \tag{17}$$

where $K$ represents the window size used for calculating the LSD and $M_k$ represents the average gray value of the pixels in the window.

## III. Experiment and Result

### A. Data and Training Details

Unlike artificial simulations that generate training data, this study employs real-world multisensor data to train the model. The Landsat series of sensors is well-known for possessing the world's longest time series of surface observation data and finds broad applications in various fields [58]. The L8 satellite has a spatial resolution of 30 m in all but one panchromatic band. In contrast, the S2 satellite is developed by ESA and is one among several satellites part of the Copernicus Earth Observation Program series. The S2 satellite constellation consists of two identical satellites, Sentinel-2A and Sentinel-2B, separated by 180° in the same orbit [59]. The S2 satellite imagery with 10 m resolution in R, G, B, and NIR bands. In our study, L8 images are utilized as the LR input data, while S2 images are used as the HR data for the surface. Finally, the Red, Green, Blue, and NIR bands from S2 were selected based on their distinct resolutions, and these selected bands are shown in Table I.

A total of 16 images worldwide were obtained for this study from the USGS and ESA websites. To minimize surface variation caused by time variation, L8 and S2 images of the same region were acquired at 10-day intervals, with the cloud cover controlled below 5%. The acquired data in this study comprises level-2 real surface reflectance products; hence, the impact of atmospheric changes is not necessary. Following the acquisition of the L8 and S2 images of the corresponding areas, we initially unified the coordinate systems of the two images and then aligned them to generate the matching LR-HR image pairs. To enable image input to the network, the LR images were cropped without overlap, producing 64 × 64 pixel patches, while the matching HR image patches were cropped to 192 × 192 pixels, resulting in a total of 25 200 image patches. After manual inspection to remove images with noise, clouds, and other interferences, the final dataset comprised 24 788 paired image patches. We will downsample the LR patches to a size of 48 × 48 for testing the model's applicability across different scale factors. Fig. 7 shows a selection of four of the images.

The PyTorch framework [60], [61] was adopted in this study to train the model, using an RTX3090 graphics card. Data augmentation was incorporated into the training process by randomly rotating and flipping the data horizontally and vertically by 90°, 180°, and 270°. The datasets were partitioned into training, validation, and test sets in an 8 : 1 : 1 ratio, respectively. The initial learning rate was set to $2e^{-4}$, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

TABLE I
BLUE, GREEN, RED, AND NIR BAND DETAILS FOR LANDSAT-8 AND SENTINEL-2

| | Landsat-8 | | | | Sentinel-2 | | | |
|---|---|---|---|---|---|---|---|---|
| | Blue | Green | Red | NIR | Blue | Green | Red | NIR |
| Wave width/μm | 0.030 | 0.057 | 0.031 | 0.021 | 0.098 | 0.031 | 0.026 | 0.115 |
| Center wave length/μm | 0.482 | 0.561 | 0.654 | 0.865 | 0.490 | 0.560 | 0.665 | 0.842 |
| Resolution/m | 30 | 30 | 30 | 30 | 10 | 10 | 10 | 10 |

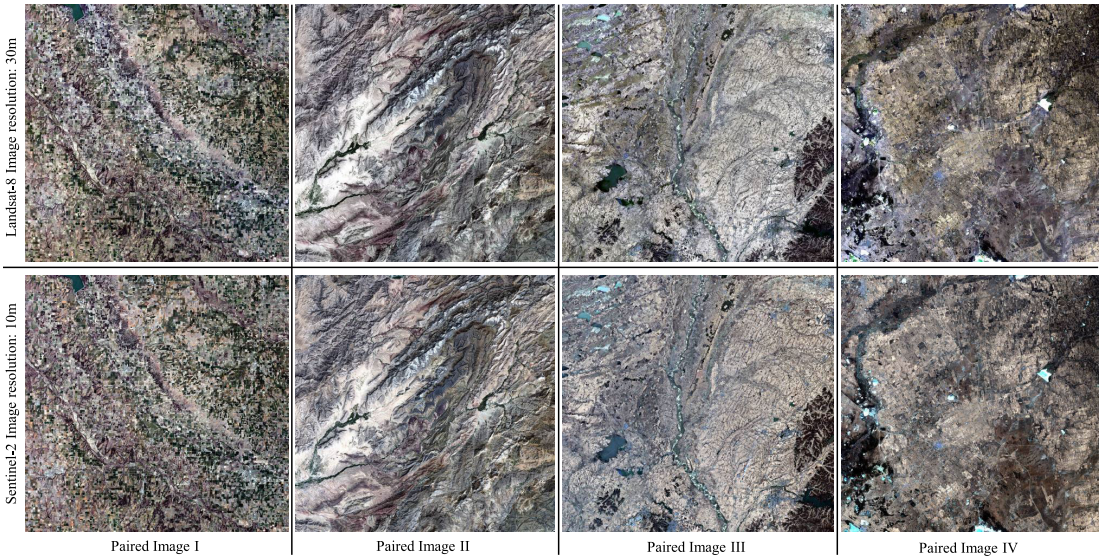



Fig. 7. Training data samples from different scenes are used in our study. The first row is 30 m LR L8 images. The second row is 10 m HR S2 images.

an Adam optimizer [62] with hyperparameters $\beta_1 = 0.9$ and $\beta_2 = 0.99$ was employed. The learning rate was reduced by a factor of 0.2 every 150 epochs, and the batch size was set to 64 for a total of 500 training epochs. The model configuration details are shown in Table II.

### B. Comparison With State-of-the-Art Methods

To demonstrate the effectiveness of the proposed model, this study compares it with several state-of-the-art deep learning-based image SR methods. SRCNN [9] is considered as the baseline method in this study. Both VDSR [63] and RCAN [17] have extremely deep network depth, and RCAN [17] introduces the attention mechanism into the SR task. Both LGCNet [14] and TransENet [23] are networks specially designed for the remote sensing image SR (RSISR). TransENet [23] and SwinIR [64] both use transformer to build the network. SRGAN [43] uses a generative adversarial architecture to train the SR network and uses the perceptual loss optimization model. Fig. 8 shows the curve of variations in PSNR and SSIM during the training process for ×3 SR. After 150 epochs, when the learning rate changes for the first time, the performance of our model starts to outperform the other models. The performance of the model does not change much in the later training, but only slightly improves at 300 and 450 epochs with the adjustment of the learning rate, which suggests that the model's proficiency has reached a plateau.

Table III and Fig. 9 display the number of parameters and average performance of various models on the test dataset. When ×3 SR, our approach outperforms the second-best model by 0.0524 dB, 0.0006, and 0.0108 in terms of PSNR, SSIM, and LPIPS metrics, respectively.

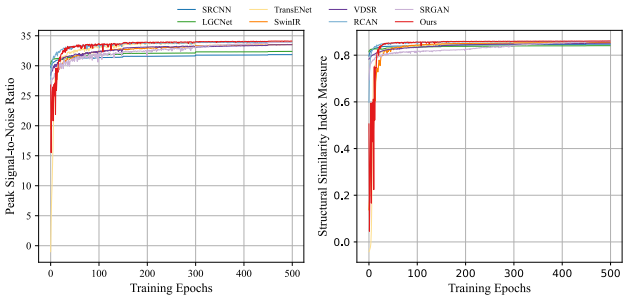



Fig. 8. Comparison of PSNR and SSIM with the number of training epochs for different models on validation data for ×3 SR.

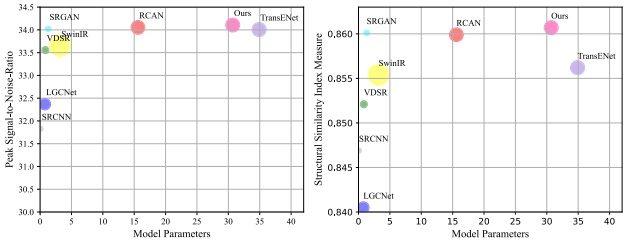



Fig. 9. Number of parameters for different models and the average performance achieved on the test dataset for ×3 SR.

When ×4 SR, TransENet achieves the highest PSNR, while our method higher the second-best model by 0.0012 and 0.0491 in SSIM and LPIPS metrics, respectively. Furthermore, we also compared the parameter count of different models. $K$ and $M$ in the parameter column of Table III stands for $1e^3$ and $1e^6$, respectively.

Figs. 10 and 11 show the comparison of the reconstruction effects of different models for RS images, which include RGB true color images and LSD images. The reconstructed image based on pixel loss optimization appears smoother when

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DETAILED CONFIGURATION OF EACH COMPONENT OF THE MODEL

| | Depth | Heads | Atrous rate | Emded dim | MLP dim |
|---|---|---|---|---|---|
| Generator | [4,4,6,5] | [3,6,12,24] | [1,2,3] | 72 | [144,288,576,1152] |
| Discriminator | 12 | 12 | - | 96 | 384 |

TABLE III
AVERAGE QUALITY EVALUATION OF DIFFERENT MODELS ON THE TEST DATASET

| Methods | Scale | PSNR | SSIM | LPIPS | SAM | MSE | Parameters |
|---|---|---|---|---|---|---|---|
| SRCNN [9] | ×3 | 31.8267 | 0.8469 | 0.3556 | 0.0475 | 0.0036 | 75.2K |
| LGCNet [14] | ×3 | 32.3676 | 0.8405 | 0.3301 | 0.0350 | 0.0033 | 769K |
| VDSR [63] | ×3 | 33.5555 | 0.8521 | 0.3264 | 0.0334 | 0.0016 | 852K |
| RCAN [17] | ×3 | 34.0542 | 0.8599 | 0.2933 | 0.0255 | 0.0032 | 15.6M |
| TransENet [23] | ×3 | 34.0065 | 0.8562 | 0.2705 | 0.0236 | 0.0014 | 34.9M |
| SwinIR [64] | ×3 | 33.6455 | 0.8554 | 0.2734 | 0.0250 | 0.0033 | 3.2M |
| SRGAN [43] | ×3 | 34.0198 | 0.8601 | 0.2673 | 0.0289 | 0.0034 | 1.3M |
| Ours | ×3 | 34.1067 | 0.8607 | 0.2565 | 0.0218 | 0.0022 | 30.7M |
| SRCNN [9] | ×4 | 28.5810 | 0.8131 | 0.4999 | 0.0958 | 0.0054 | 75.2K |
| LGCNet [14] | ×4 | 30.7673 | 0.8285 | 0.4467 | 0.0546 | 0.0047 | 769K |
| VDSR [63] | ×4 | 31.6063 | 0.8387 | 0.4104 | 0.0395 | 0.0037 | 852K |
| RCAN [17] | ×4 | 32.2842 | 0.8536 | 0.4344 | 0.0916 | 0.0042 | 15.6M |
| TransENet [23] | ×4 | 32.4972 | 0.8422 | 0.3971 | 0.0940 | 0.0028 | 34.9M |
| SwinIR [64] | ×4 | 32.3861 | 0.8555 | 0.4441 | 0.0929 | 0.0045 | 3.2M |
| SRGAN [43] | ×4 | 32.3232 | 0.8367 | 0.3526 | 0.0921 | 0.0043 | 1.3M |
| Ours | ×4 | 32.4679 | 0.8567 | 0.3035 | 0.0347 | 0.0031 | 30.7M |

compared to the RGB image. However, its reconstruction effect is blurrier at the edges. In contrast, the method proposed in this study shows the clearest reconstruction result and closely resembles the HR image in terms of capturing details. To assess the accuracy of the reconstruction, we calculate the LSD in $5 \times 5$ window and obtain the corresponding standard deviation map. This analysis demonstrates that the proposed method is capable of reconstructing detailed information that closely resembles the the real image.

Fig. 12 shows the reconstruction effect of MSWAGAN for each band. The results demonstrate that the approach presented in this article effectively enhances the spatial resolution of each band and reconstructs more information. Additionally, it becomes apparent that the NIR band contains a greater amount of feature information compared to the R, G, and B bands. For instance, the circle texture on the disk is clearly visible in the NIR band but not in the other three bands. This also verifies the necessity of SR for multispectral remote sensing images and the wide application value of the research in this article.

Due to differences in imaging wavelengths and sensors, there exists a certain bias between the spectral profiles of L8 and S2 images. During the training process, the model implicitly learns the spectral mapping relationship between L8 and S2 images. Fig. 13 illustrates the changes in the spectra of several typical land surfaces before and after SR. It can be observed that, compared to LR images, SR images exhibit spectra more similar to HR images across various bands.

## IV. DISCUSSION

### A. Multiscale Sliding Window Attention

MSWA is the new attention mechanism proposed in this study. The computer of MSWA with sliding windows of different atrous rates to extract local multiscale complex features of RS images without increasing the network parameters. This study explores the effect of MHA and MSWA on the results by incorporating three different attention mechanisms. The results of MWSA ablation experiments are shown in Table IV. Initially, we evaluated the network performance with only MHA, which computes self-attention at a global scale without focusing on multiscale information. This configuration yielded the poorest results in the evaluation metrics. Subsequently, we tested the performance of the network with only MSWA, and the results indicated that the utilization of multiscale information enhanced the quality of the network reconstructed images by 0.0273dB and 0.0008 in PSNR and SSIM metrics, respectively, compared to using only MHA. Given that MSWA lacks global attention but introduces local and multiscale information into the network, we combined MHA and MSWA once again. Specifically, MSWA was employed to learn local multiscale information, while MHA was leveraged to learn global information about the image. By combining multiscale and global information, the reconstruction quality of images was further improved compared to using a single attention mechanism. It is noteworthy that MSWA computes attention within a local window, resulting in the same number of parameters as MHA, but with a faster inference speed.

TABLE IV
ABLATION EXPERIMENT OF MSWA FOR ×3 SR

| MSWA Stage | VIT Stage | PSNR | SSIM | LPIPS | SAM | MSE |
|---|---|---|---|---|---|---|
| 0 | 4 | 33.9964 | 0.8561 | 0.2794 | 0.0314 | 0.0032 |
| 0 | 1 | 33.3625 | 0.8397 | 0.3513 | 0.0812 | 0.0041 |
| 4 | 0 | 34.0237 | 0.8569 | 0.2846 | 0.0298 | 0.0036 |
| 1 | 0 | 33.4357 | 0.8417 | 0.3406 | 0.0756 | 0.0039 |
| 2 | 2 | 34.0821 | 0.8607 | 0.2569 | 0.0224 | 0.0026 |

### B. Loss Function

The loss function plays a crucial role in updating the network weights, and various loss functions exhibit distinct

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

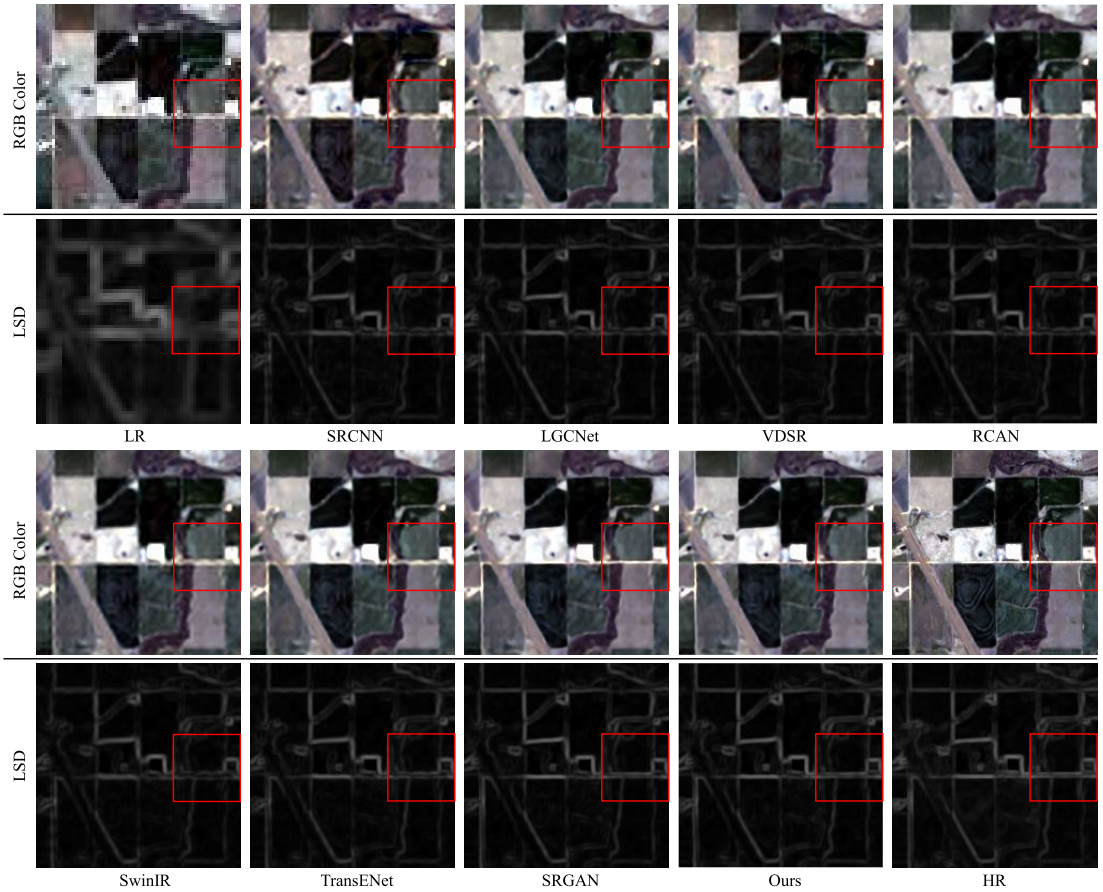



Fig. 10. Comparison of visual reconstruction effects of different models for ×3 SR.

TABLE V
ABLATION EXPERIMENT OF LOSS FUNCTION FOR ×3 SR

| Loss Function | PSNR | SSIM | LPIPS | SAM | MSE |
|---|---|---|---|---|---|
| $\mathcal{L}_{\text{Pixel}}$ | 34.1028 | 0.8559 | 0.2613 | 0.0229 | 0.0021 |
| $\mathcal{L}_{\text{Pixel}} + \mathcal{L}_{\text{vgg}}$ | 34.0149 | 0.8603 | 0.2572 | 0.0234 | 0.0033 |
| $\mathcal{L}_{\text{Pixel}} + \mathcal{L}_{\text{RaGAN}}$ | 34.0751 | 0.8595 | 0.2581 | 0.0241 | 0.0024 |
| $\mathcal{L}_{\text{Pixel}} + \mathcal{L}_{\text{vgg}} + \mathcal{L}_{\text{RaGAN}}$ | 34.0821 | 0.8607 | 0.2569 | 0.0224 | 0.0026 |

TABLE VI
ABLATION EXPERIMENT OF MODEL DEPTH FOR ×3 SR

| Depth | PSNR | SSIM | LPIPS | SAM | MSE | Parameters |
|---|---|---|---|---|---|---|
| [6, 6, 8, 5] | 34.0821 | 0.8607 | 0.2569 | 0.0224 | 0.0026 | 65.4M |
| [4, 4, 6, 5] | 34.1067 | 0.8607 | 0.2565 | 0.0218 | 0.0022 | 30.7M |
| [2, 2, 4, 4] | 34.0238 | 0.8602 | 0.2573 | 0.0231 | 0.0031 | 27.2M |

characteristics that can impact the results of image reconstruction in different ways. In this study, four experiments were conducted to assess the impact of different loss functions on the quality of RS image reconstruction. As shown in Table V, the results of our loss function ablation experiments demonstrate that optimizing the network solely based on pixel loss leads to an improvement in SR image PSNR, albeit it yields the lowest value in terms of the SSIM metric. This can be attributed to the fact that pixel loss utilizes PSNR as the optimization objective instead of focusing on perceptual distance.

### C. Model Depth

In the field of deep learning, researchers have long pursued the goal of building extremely deep networks. Generally, deeper networks can improve learning ability; however, it is not always the case that the deeper the network, the better the performance. In reality, a deeper network can result in overfitting and reduced model stability. To explore the effect of network depth, model depth ablation experiments were conducted, and the results are presented in Table VI. In Table IV, the first two numbers in the "depth" column denote the quantity of MSWA blocks within the MSWA stage, while the corresponding last two numbers denote the count of VIT blocks in the VIT stage. In this study, we constructed three different models with varying numbers of MSWA blocks and VIT blocks in the stage. The best results were obtained by the MSWAGAN with a depth of [4, 4, 6, 5]. This finding indicates that increasing the model depth may lead to overfitting issues, which reduce model effectiveness. In addition, the number of network parameters increases exponentially with depth, necessitating more computational resources.

### D. Patch Size

Before feeding shallow features into the deep feature extraction module, it is necessary to partition the shallow features into $P \times P$ patches. This process is achieved using a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

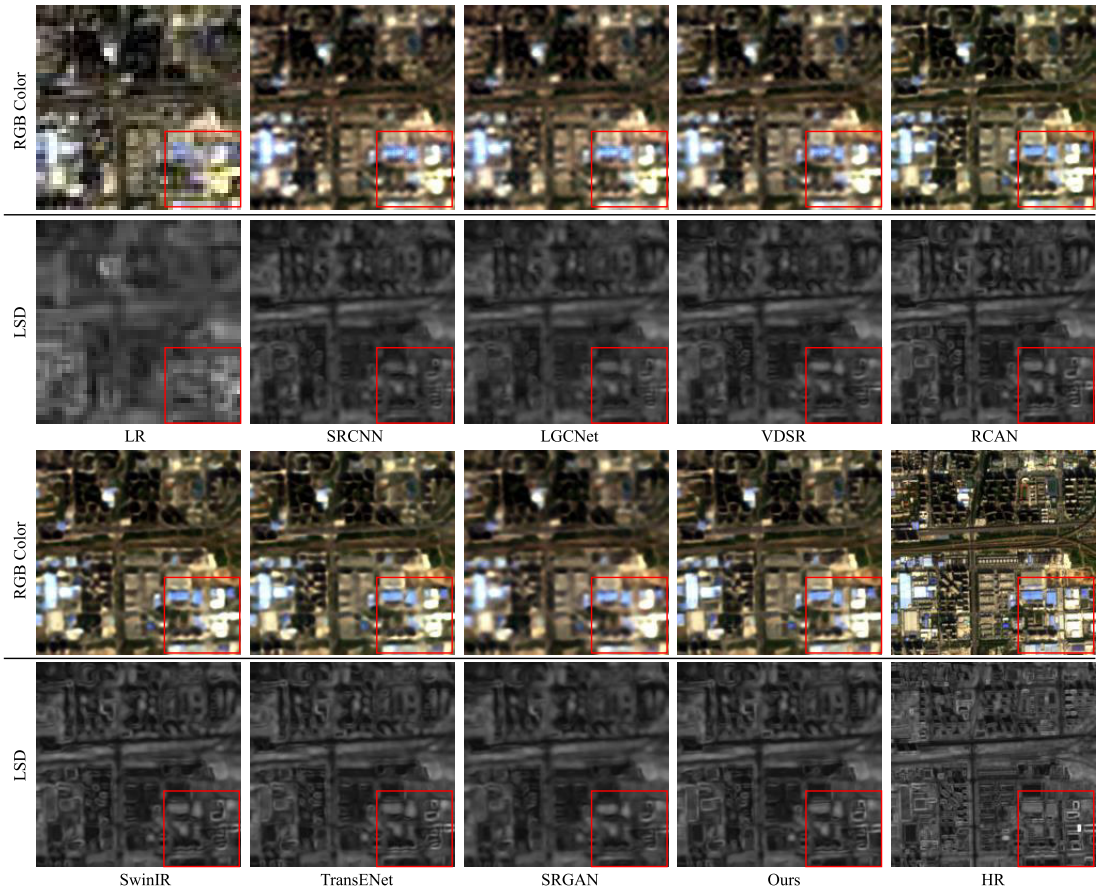



Fig. 11. Comparison of visual reconstruction effects of different models for $\times 4$ SR.

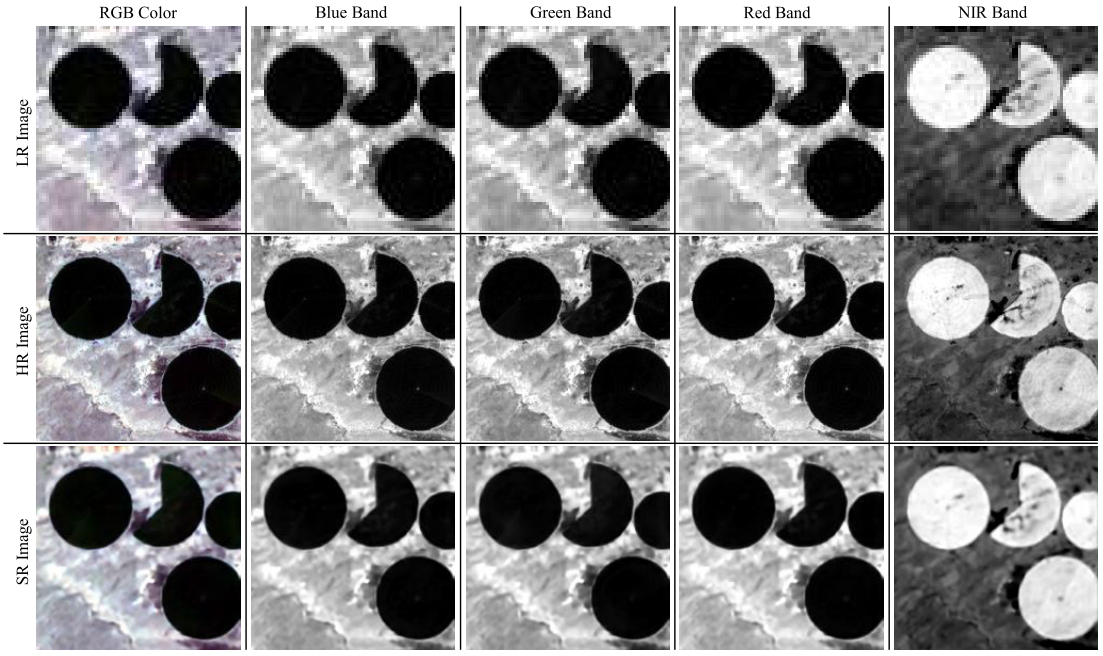



Fig. 12. Reconstruction effect of our method on each band of the image for $\times 3$ SR.

convolutional layer with a kernel size equal to $P$, essentially acting as a form of downsampling. The size of $P$ directly influences the reconstruction performance of the model. We explored the impact of different values of $P$ on $\times 4$ SR effects, and the results are presented in Table VII. It can be observed that the model's performance decreases with an increase in $P$. A larger $P$ implies more information loss, as SR is an ill-posed problem of reconstructing high-frequency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

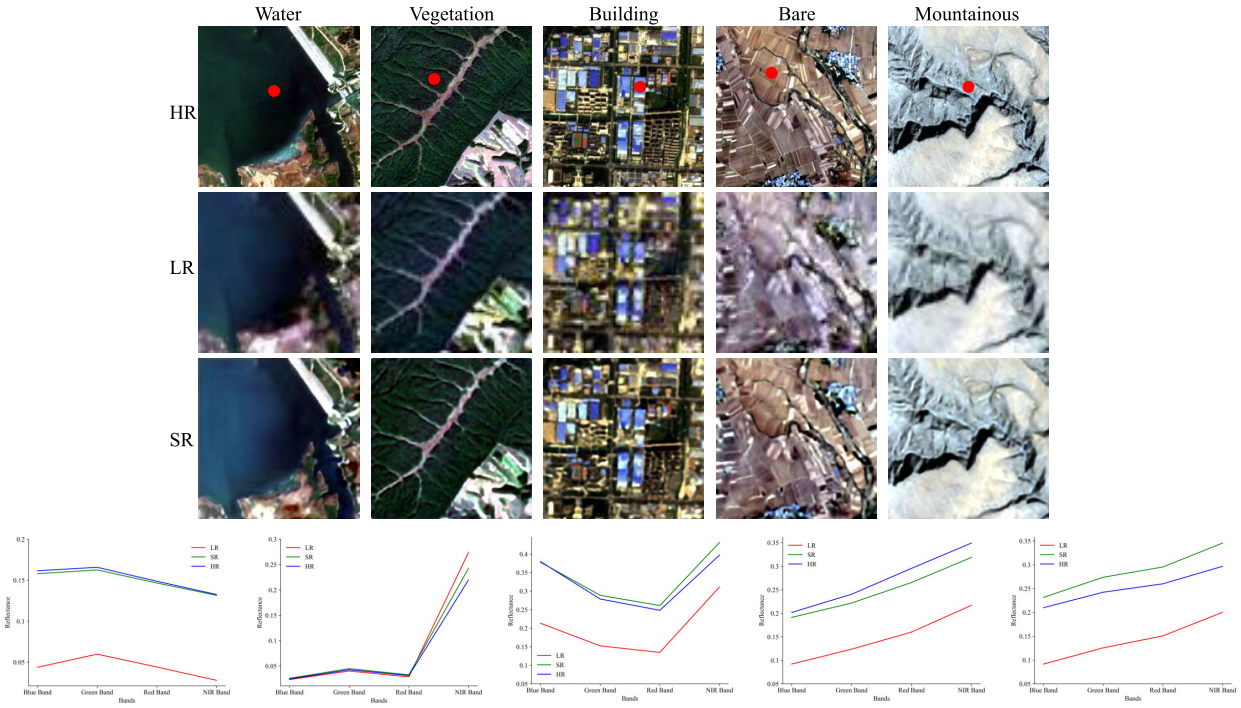



Fig. 13. Spectral curve variations of distinct land surfaces before and after SR within the marked red circular regions.

TABLE VII
ABLATION EXPERIMENT OF $P$ FOR $\times 4$ SR

| $P \times P$ | PSNR | SSIM | LPIPS | SAM | MSE |
|---|---|---|---|---|---|
| $4 \times 4$ | 32.4679 | 0.8567 | 0.3035 | 0.0347 | 0.0031 |
| $6 \times 6$ | 32.3251 | 0.8523 | 0.3467 | 0.0392 | 0.0038 |
| $8 \times 8$ | 31.8232 | 0.8391 | 0.3373 | 0.0359 | 0.0044 |

information from an LR image, and increased information loss leads to poorer reconstruction results. Meanwhile, a smaller $P$ entails higher computational overhead, considering that the computational complexity of the transformer is quadratic in relation to the image size. Therefore, a balanced consideration between computational resources and model performance is required.

### *E. SR Results in Different Scenarios*

Remote sensing images typically contain rich features, and the reconstruction effects of SR models tend to vary across different scenes. In this study, we examine the reconstruction effects of different models on several typical scenes. Fig. 14 depicts the reconstruction effects of a town building scene. The building scene features are relatively intricate, and even the S2 image has ambiguous edge features that appear as blurred, colored highlights. The SRCNN [9] and LGCNet [14] models have a limited number of network layers and are therefore less capable of fitting complex scenes. The reconstructed details are thus limited, and the spectral tend to shift. The proposed method in this study reconstructs more details compared to other methods; however, the reconstructed image still falls short of the real ground image.

Fig. 15 illustrates the reconstruction outcomes of a mountainous area characterized by a vast region of smooth and homogenous terrain devoid of intricate edge textures. The difference in reconstruction effects across different models is not apparent in this scene. This indicates that when the scene is relatively straightforward, lacking excessively complex information and sharp edges, it is easier to achieve improved reconstruction output compared to complex and demanding scenes.

Fig. 16 displays the reconstruction results of a scene containing sharp edges. The proposed method in this study obtained better reconstruction results on the edges compared to other methods. The edge reconstruction effect based on the CNN method appears visually blurred, with adhesion effects between adjacent lines. In contrast, the reconstruction effect of the proposed method is clearer, enabling easier distinction between different lines without any adhesion. In addition, it successfully reconstructs the raised lines on the smooth area on the right side. These outcomes indicate that the proposed method outperforms the other methods used for comparison.

### *F. Problems With This Study and Future Work*

The aim of this study is to explore the potential use of SR techniques for remote sensing imagery to generate high-quality data. To achieve this, we utilized a real-world L8 image training model to enhance the resolution from 30 to 10 m in R, G, B, and NIR bands. During multispectral image SR, one of the critical issues is constraining spectral changes. This study did not consider any spectral constraints. MSWA can implicitly and adaptively learn spectral features, but there are still observable spectral differences between the reconstructed images and the HR images. Therefore, we expect to build a joint spatial-spectral loss aimed at optimizing the spectral information of LR images to achieve joint spatial-spectral SR.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

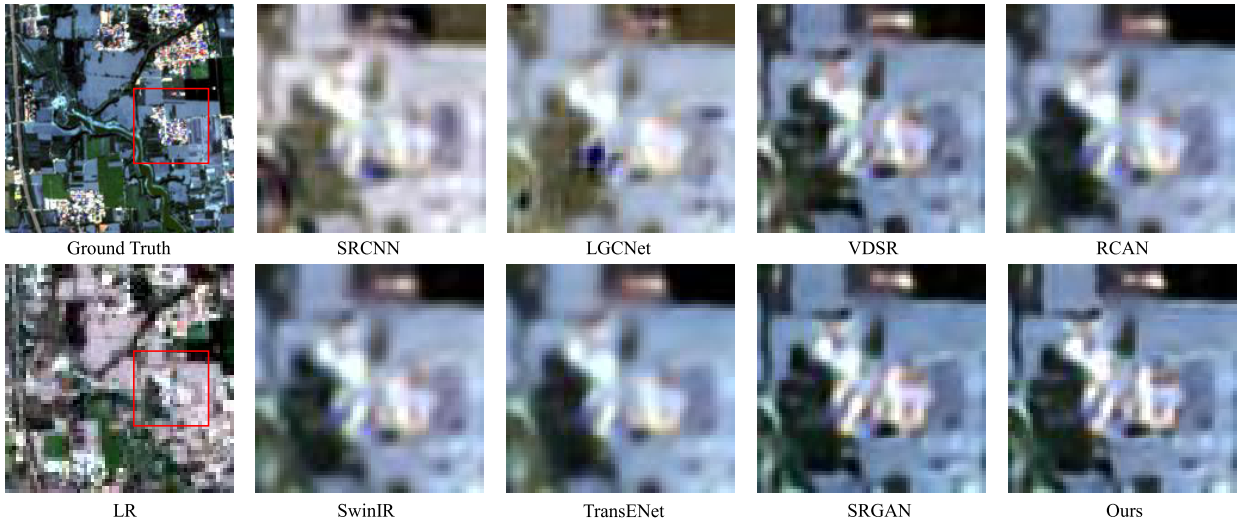



Fig. 14. Reconstruction effect of different models on complex town scenes for $\times 3$ SR.

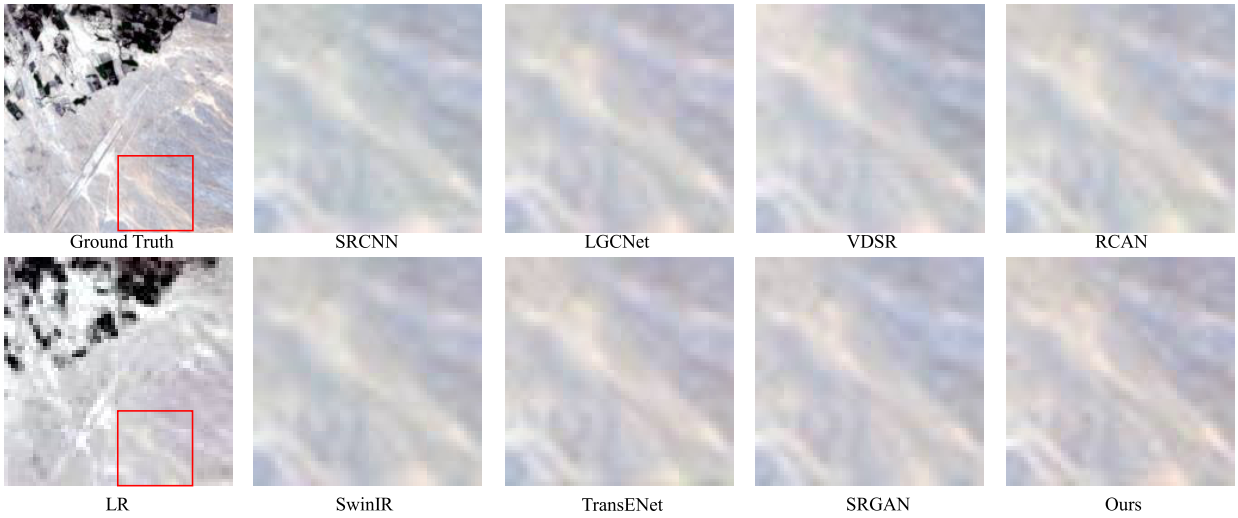



Fig. 15. Reconstruction effect of different models on mountain smooth areas for $\times 3$ SR.

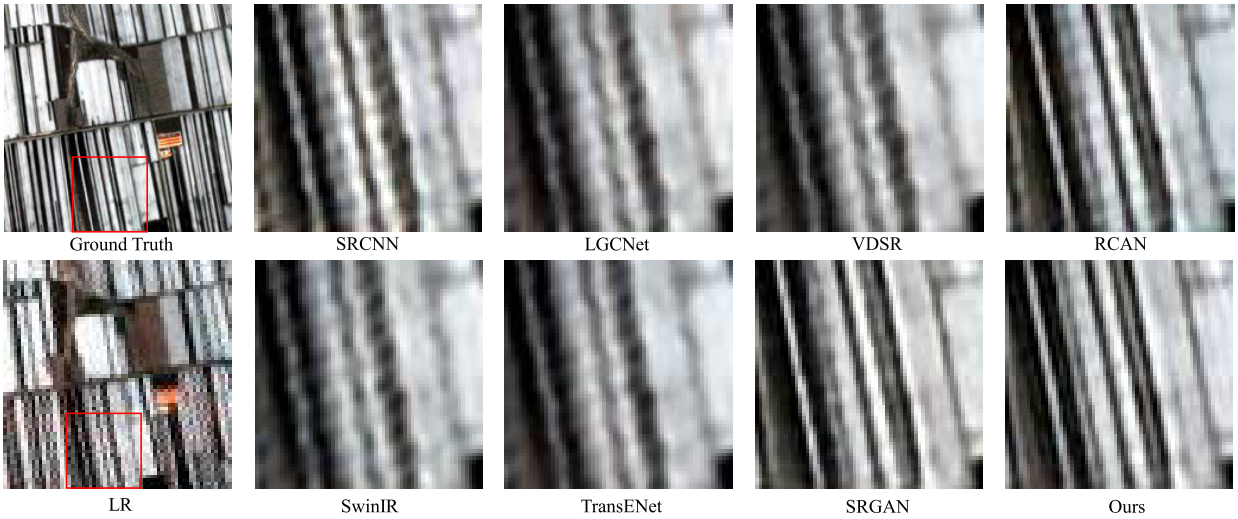



Fig. 16. Reconstruction effect of different models on sharp lines for $\times 3$ SR.

The L8 images used in this study cover a vast area globally, and the amount of data is enormous. However, current hardware conditions do not support extensive-scale training. In the future, we plan to target specific regions with research value for SR processing to develop an SR image dataset. Our plan is to carry out various remote sensing applications based on this dataset, such as land-use classification, target detection, and crop extraction, among others. We expect to obtain highly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

accurate results from these applications. Furthermore, we plan to apply SR on the entire Landsat dataset.

## V. Conclusion

In this study, we propose a two-stage GAN that combines the advantages of CNN and transformers to address the challenges encountered in SR methods when dealing with remote sensing images. These challenges include the complexity of the features present in such images, which can be difficult to learn, and the loss of long-term interpixel correlations. The proposed solution consists of two stages. In the first stage, we employ a newly introduced attention mechanism called MSWA to capture local multiscale features of RS images. In the second stage, we utilize a standard MHA mechanism to extract global features and model the long-term interpixel correlations. Furthermore, we have curated a SR dataset of multispectral remote sensing images that includes 16 L8 and S2 images from various locations around the world. Through a series of experiments conducted on the generated imagery, the results demonstrate that our proposed method enhances the resolution of multispectral RS images and achieves superior reconstruction of remote sensing images compared to other methods. The high-quality SR dataset provides a valuable resource for remote sensing applications and further caters to the demands of high spatial resolution remote sensing image data across various fields.

## References


[1] S. K. Seelan, S. Laguette, G. M. Casady, and G. A. Seielstad, "Remote sensing applications for precision agriculture: A learning community approach," *Remote Sens. Environ.*, vol. 88, nos. 1–2, pp. 157–169, Nov. 2003.

[2] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[3] J. A. Voogt and T. R. Oke, "Thermal remote sensing of urban climates," *Remote Sens. Environ.*, vol. 86, no. 3, pp. 370–384, Aug. 2003.

[4] T. P. P. Sharma, J. Zhang, U. A. Koju, S. Zhang, Y. Bai, and M. K. Suwal, "Review of flood disaster studies in Nepal: A remote sensing perspective," *Int. J. Disaster Risk Reduction*, vol. 34, pp. 18–27, Mar. 2019.

[5] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2014.

[6] L. Zhang and X. Wu, "An edge-guided image interpolation algorithm via directional filtering and data fusion," *IEEE Trans. Image Process.*, vol. 15, no. 8, pp. 2226–2238, Aug. 2006.

[7] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[8] H. Stark and P. Oskoui, "High-resolution image recovery from image-plane arrays, using convex projections," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 6, no. 11, pp. 1715–1726, Nov. 1989.

[9] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[10] L. Liebel and M. Körner, "Single-image super resolution for multispectral remote sensing data using convolutional neural networks," *ISPRS-Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 883–890, Jun. 2016.

[11] T. Tong, G. Li, X. Liu, and Q. Gao, "Image super-resolution using dense skip connections," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4809–4817.

[12] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.

[13] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[14] S. Lei, Z. Shi, and Z. Zou, "Super-resolution for remote sensing images via local-global combined network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1243–1247, Aug. 2017.

[15] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5835–5843.

[16] J. Li, F. Fang, K. Mei, and G. Zhang, "Multi-scale residual network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds. Cham, Switzerland: Springer, Sep. 2018, pp. 527–542.

[17] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Oct. 2018, pp. 294–310.

[18] D. Zhang, J. Shao, X. Li, and H. T. Shen, "Remote sensing image super-resolution via mixed high-order attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5183–5196, Jun. 2021.

[19] B. Niu et al., "Single image super-resolution via a holistic attention network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds. Cham, Switzerland: Springer, 2020, pp. 191–207.

[20] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30. Red Hook, NY, USA: Curran Associates, 2017, pp. 1–9.

[21] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Oct. 2021, pp. 1–21.

[22] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5790–5799.

[23] S. Lei, Z. Shi, and W. Mo, "Transformer-based multistage enhancement for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5615611, doi: 10.1109/TGRS.2021.3136190.

[24] M. V. Conde, U.-J. Choi, M. Burchi, and R. Timofte, "Swin2SR: SwinV2 Transformer for compressed image super-resolution and restoration," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, 2022, pp. 669–687.

[25] J. Tu, G. Mei, Z. Ma, and F. Piccialli, "SWCGAN: Generative adversarial network combining Swin transformer and CNN for remote sensing image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5662–5673, 2022.

[26] T. Lu, J. Wang, Y. Zhang, Z. Wang, and J. Jiang, "Satellite image super-resolution via multi-scale residual deep neural network," *Remote Sens.*, vol. 11, no. 13, p. 1588, Jul. 2019.

[27] S. Lei and Z. Shi, "Hybrid-scale self-similarity exploitation for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022, Art. no. 5401410, doi: 10.1109/TGRS.2021.3069889.

[28] X. Dong, L. Wang, X. Sun, X. Jia, L. Gao, and B. Zhang, "Remote sensing image super-resolution using second-order multi-scale networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3473–3485, Apr. 2021.

[29] Y. Lee, J. Kim, J. Willette, and S. J. Hwang, "MPViT: Multi-path vision transformer for dense prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7277–7286.

[30] W. Wang et al., "CrossFormer: A versatile vision transformer hinging on cross-scale attention," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–15.

[31] H. Fan et al., "Multiscale vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6824–6835.

[32] S. Ren, D. Zhou, S. He, J. Feng, and X. Wang, "Shunted self-attention via multi-scale token aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10853–10862.

[33] Y. Chen et al., "Mobile-Former: Bridging MobileNet and transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5260–5269.

[34] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 367–376.

[35] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016, pp. 1–13.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.

[36] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," 2020, *arXiv:2004.05150*.

[37] A. Hassani, S. Walton, J. Li, S. Li, and H. Shi, "Neighborhood attention transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6185–6194.

[38] G. Li, D. Xu, X. Cheng, L. Si, and C. Zheng, "SimViT: Exploring a simple vision transformer with sliding windows," in *Proc. IEEE Int. Conf. Multimedia Expo. (ICME)*, Jul. 2022, pp. 1–6.

[39] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," 2021, *arXiv:2103.14030*.

[40] I. J. Goodfellow et al., "Generative adversarial networks," 2014, *arXiv:1406.2661*.

[41] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5967–5976.

[42] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.

[43] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 105–114.

[44] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Y. Bengio and Y. LeCun, Eds. May 2015, pp. 1–14.

[45] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, 2018, pp. 63–79.

[46] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.

[47] S. Jia, Z. Wang, Q. Li, X. Jia, and M. Xu, "Multiattention generative adversarial network for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5624715, doi: 10.1109/TGRS.2022.3180068.

[48] R. Dong, L. Zhang, and H. Fu, "RRSGAN: Reference-based super-resolution for remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601117.

[49] S. Lei, Z. Shi, and Z. Zou, "Coupled adversarial training for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3633–3643, May 2020.

[50] X. Chu, Z. Tian, B. Zhang, X. Wang, and C. Shen, "Conditional positional encodings for vision transformers," 2021, *arXiv:2102.10882*.

[51] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.

[52] W. Shi et al., "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1874–1883.

[53] A. Jolicoeur-Martineau, "The relativistic discriminator: A key element missing from standard GAN," 2018, *arXiv:1807.00734*.

[54] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.

[55] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.

[56] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.

[57] F. A. Kruse et al., "The spectral image processing system (SIPS)—Interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, pp. 145–163, Jun. 1993.

[58] USGS. *Landsat Missions*. Accessed: Jan. 12, 2023. [Online]. Available: https://earthexplorer.usgs.gov/

[59] ESA. *Sentinel-2 User Handbook*. Accessed: Jul. 24, 2015. [Online]. Available: https://sentinels.copernicus.eu/documents/247904/685211/Sentinel-2_User_Handbook

[60] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," 2019, *arXiv:1912.01703*.

[61] A. Rogozhnikov, "EINOPS: Clear and reliable tensor manipulations with Einstein-like notation," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–21.

[62] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015, pp. 1–15.

[63] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[64] J. Liang, J. Cao, G. Sun, K. Zhang, L. van Gool, and R. Timofte, "SwinIR: Image restoration using Swin Transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.



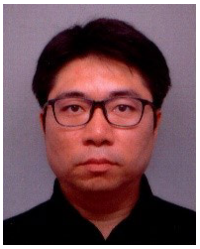
**ChunYang Wang** received the Ph.D. degree from the School of Surveying and Land Information Engineering, Henan Polytechnic University, Jiaozuo, China, in 2015.

He spent one year from 2018 to 2019 as a Visiting Scholar at Chiba University, Chiba, Japan. He is currently an Associated Professor with the School of Computer Science and Technology, Henan Polytechnic University. His research interests include image processing, artificial intelligence and deep learning remote sensing applications, and spatial and temporal big data remote sensing applications.



**Xian Zhang** received the B.S. degree from Henan Polytechnic University, Jiaozuo, China, in 2021, where he is currently pursuing the M.S. degree with the School of Surveying and Land Information Engineering.

His research interests include deep learning, remote sensing image super-resolution, and remote sensing image segmentation.



**Wei Yang** received the B.S. and M.S. degrees in physics and civil engineering from Beijing Normal University, Beijing, China, in 2005 and 2008, respectively, and the Ph.D. degree in integrative environmental science from the University of Tsukuba, Tsukuba, Japan, in 2011.

He is currently an Assistant Professor with the Center for Environmental Remote Sensing, Chiba University, Chiba, Japan, where he teaches and conducts research on quantitative remote sensing and ecosystem ecology. His research interests include satellite observations of terrestrial and aquatic ecosystems, as well as their application to ecological modeling.



**GaiGe Wang** (Member, IEEE) is currently a Professor with the Ocean University of China, Qingdao, China. His entire publications have been cited over 10 000 times (Google Scholar). The latest Google H-index and i10-index are 58 and 109, respectively; 15 and 66 papers are selected as Highly Cited Paper by Web of Science and Scopus, respectively. His research interests are swarm intelligence, evolutionary computation, and big data optimization.

Prof. Wang is a Senior Member of SAISE and SCIEI and a member of IEEE CIS and ISMOST. He was selected as one of "2021 Highly Cited Researchers" by Clarivate. He was selected as one of "2021 and 2020 Highly Cited Chinese Researchers" in computer science and technology by Elsevier. He was selected as one of "MDPI 2021 Most Influential Author Award" in computer science and mathematics by MDPI. He was selected as the World's Top 2% Scientists in 2020. One of his papers was selected as the "100 Most Influential International Academic Papers in China," one of his papers ranks 1 in the selection of the latest high-impact publications in computer science by Chinese researchers from Springer Nature in 2019. He served as the Editors-in-Chief for the *International Journal of Artificial Intelligence and Soft Computing*, an Early Career Advisory Board Member for IEEE/CAA JOURNAL OF AUTOMATICA SINICA (SCI), an Editorial Advisory Board Member for *Communications in Computational and Applied Mathematics* (CCAM), an Associate Editor for *IJCISIM*, an Editorial Board Member for the *Journal of Computational Design and Engineering*, *Mathematics*, *IJBIC*, the *Karbala International Journal of Modern Science*, and the *Journal of AIS*. He served as a Guest Editor for many journals including *Mathematics*, *IJBIC*, *FGCS*, *Memetic Computing*, and *Operational Research*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.



**XingWang Li** (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees from the University of Electronic Science and Technology of China, Chengdu, China, and the Beijing University of Posts and Telecommunications, Beijing, China, in 2010 and 2015, respectively.

From 2010 to 2012, he was with Comba Telecom Ltd., Guangzhou, China, as an Engineer. He spent one year from 2017 to 2018 as a Visiting Scholar at Queen's University Belfast, Belfast, U.K. He is currently an Associated Professor with the School of Physics and Electronic Information Engineering, Henan Polytechnic University, Jiaozuo, China. His research interests span wireless communications, intelligent transport systems, artificial intelligence, and the Internet of Things.

Dr. Li was a recipient of Exemplary Reviewer of IEEE TRANSACTIONS ON COMMUNICATIONS and the *Journal of Electronics and Information Technology* in 2022. He is on the Editorial Board of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE SYSTEMS JOURNAL, IEEE SENSORS JOURNAL, and *Physical Communication*. He served as a Guest Editor for the Special Issue on "Integrated Sensing and Communications (ISAC) for 6G IoE" of IEEE INTERNET OF THINGS JOURNAL, "Computational Intelligence and Advanced Learning for Next-Generation Industrial IoT" of IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, and "AI-Driven Internet of Medical Things for Smart Healthcare Applications: Challenges, and Future Trends" of IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS. He served as a TPC Member for IEEE ICC and IEEE GLOBECOM.



**JianLong Wang** received the B.S. degree in computer science and technology from Zhengzhou University, Zhengzhou, China, in 2012, the M.S. degree in computer science and technology from Henan Polytechnic University, Jiaozuo, China, in 2015, and the Ph.D. degree from Xidian University, Xi'an, China, in 2021.

He is currently a Lecturer with the School of Computer Science and Technology, Henan Polytechnic University. His research interests include image processing, sparse representation, machine learning, deep learning, and polarimetric synthetic aperture radar image classification.



**BiBo Lu** received the B.S. and M.S. degrees in computational math and the Ph.D. degree in applied math from Jilin University, Changchun, China, in 2002, 2005, and 2008, respectively.

He is currently a Professor with the School of Computer Science and Technology, Henan Polytechnic University, Jiaozuo, China. His research interests include image processing, video processing, and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:45:00 UTC from IEEE Xplore. Restrictions apply.